Muhammad Zaman

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Professional Summary

University Lecturer and Ph.D. researcher in Computer Science with more than five years of higher-education teaching experience. Expertise includes Artificial Intelligence, Machine Learning, Deep Learning, Computer Vision, Image Processing, programming, and research methods. Experienced in outcome-based education (OBE), curriculum and assessment design, postgraduate advising, thesis supervision, final-year project supervision, and academic quality assurance. Published in peer-reviewed IEEE, ACM, Springer, and Elsevier venues and actively supports students from research problem formulation through scholarly publication.

Core Competencies

University Teaching • Curriculum Development • Outcome-Based Education (OBE) • Course and Laboratory Design • Student Assessment • Academic Advising • Postgraduate Supervision • Research Mentoring • Artificial Intelligence • Machine Learning • Deep Learning • Computer Vision • Image Processing • Python • C/C++

Education

2025 – 2028 (Expected): Doctor of Philosophy (Ph.D.) in Computer Science

The Superior University, Lahore, Pakistan
Research Supervisor: Prof. Dr. Muhammad Waseem Iqbal
GPA: 3.80/4.00
Credit Hours Completed: 26
Coursework: Advanced Research Methods; Advanced Machine Learning; Advanced Natural Language Processing; Advanced Image Processing; Advanced Data Science; Advanced Computer Vision; Fehm-e-Quran I; Fehm-e-Quran II

2021 – 2023: Master of Science in Computer Science (MSCS)

COMSATS University Islamabad, Islamabad, Pakistan.
Supervised by Prof. Dr. Majid Iqbal
GPA: 3.22/4.00
Total CP: 80.55
Total Credit Hour: 31
Pre-requisite: Research Methodology
Core Coursework: Advanced Algorithm Analysis, Advanced Operating Systems, Advanced Computer Architecture, Theory of Programming Languages
Relevant Coursework: Artificial Intelligence, Machine Learning, Deep Learning, Computer Vision
Thesis: Circuit Based Error Detection and Correction Scheme for Quantum System

2017 – 2019: Master in Computer Science (MCS)

Islamia University, Bahawalpur, Pakistan
GPA: 3.11/4.00
Total CP: 72.58
Total Credit Hour: 72
Thesis: Intelligent System for Automated Medical Image Diagnostics

Experience

October 2024 – Present: Lecturer

Superior University, Lahore, Pakistan
Department Department of Computer Science & IT
Email: m.zaman.sgd@superior.edu.pk
- Deliver research-driven teaching in **AI, Machine Learning, Computer Vision, Image Processing, and**

Operating Systems at undergraduate and postgraduate levels.
- Teach **Programming and Research Methods**, emphasizing algorithmic thinking, statistical analysis, and academic writing.
- Integrate **AI frameworks, simulation tools, and real-world datasets** to bridge theory and practical implementation.
- Design and implement **project-based, lab, and research-oriented assessments** to enhance analytical skills.
- Supervise **MS theses and undergraduate FYPs**, guiding end-to-end research from problem formulation to publication.
- Mentor students in **AI and intelligent systems**, supporting high-quality **journal and conference publications**.
- Monitor research progress to ensure **quality, timelines, and academic standards**.
- Serve as **Postgraduate Coordinator and Graduate Advisor**, overseeing academic planning and student development.
- Serve as **Student Voice Lead (SVL) Advisor** for the Department of Computer Science & IT, supporting structured student representation, communication, feedback, and coordination with departmental leadership.
- Lead **thesis defenses and research committees**, ensuring rigorous evaluation and research governance.
- Foster **interdisciplinary research and innovation**, contributing to academic leadership and institutional initiatives.

October 2023 – September 2024: Lecturer
University of Lahore, Lahore, Pakistan
Department Department of Computer Science
Email: muhammad.zaman@cs.uol.edu.pk
- Delivered undergraduate and postgraduate instruction in **AI, Machine Learning, Computer Vision, Algorithms, and Programming Languages**, integrating theory with practical implementation.
- Designed course delivery emphasizing **analytical reasoning, algorithmic optimization, and computational complexity**.
- Integrated **real-world datasets, case studies, and project-based learning** to bridge theory and applied AI problem-solving.
- Taught and mentored **BSCS and MSCS students**, fostering critical thinking, independent research, and scientific reasoning.
- Developed and refined curricula, labs, and assessments aligned with **OBE standards and international benchmarks**.
- Served as **Batch Head**, overseeing student progression, academic advising, and faculty–student coordination.
- Implemented advising systems to improve **student retention, performance, and timely graduation**.
- Contributed to **research and academic committees**, including proposal evaluation and curriculum planning.
- Participated in **admissions, scheduling, and academic event management**, supporting institutional operations.
- Promoted a **research-driven academic culture**, encouraging innovation, interdisciplinary collaboration, and student engagement.

July 2020 – July 2023: Lecturer
Falah Institute of Nursing & Allied Health Sciences, Rawalpindi, Pakistan

- Delivered undergraduate instruction in **Programming, Operating Systems, Web Development, and Databases**, integrating theory with practical implementation.
- Applied pedagogical approaches emphasizing **computational thinking, algorithm design, and software engineering principles**.
- Developed student competencies in **programming, system design, and software architecture** aligned with industry standards.
- Designed and updated curricula, labs, and assessments in line with **OBE frameworks and emerging technologies**.
- Created **hands-on lab environments and project-based learning** to support experiential education.
- Developed comprehensive assessment systems including **assignments, projects, and examinations**.
- Supervised and mentored student projects, enhancing **problem-solving, debugging, and technical**

skills.
- Provided continuous academic guidance to improve **coding practices and software design proficiency**.
- Contributed to institutional roles including **QEC, ORIC, and Admissions**, supporting quality assurance and research activities.
- Supported academic leadership as **Assistant to the Dean**, contributing to planning, coordination, and institutional development.

Courses Taught

Artificial Intelligence and Data Science: Artificial Intelligence, Introduction to Machine Learning, Machine Learning, Computer Vision, Digital Image Processing

Systems and Theory: Advanced Operating Systems, Advanced Algorithm Analysis, Theory of Programming Languages

Programming and Software Development: Programming Fundamentals (C++), Object-Oriented Programming (C++), Web Development, Database Management Systems

General and Research-Oriented Courses: Introduction to ICT, Research Methods

Technical Skills and Competencies

- Expertise in **Computer Vision and Image Processing**, including segmentation, object detection, feature extraction, and pattern recognition using deep learning.
- Strong proficiency in **Machine Learning and Deep Learning**, covering model design, optimization, evaluation, and scalable AI solutions.
- Advanced **Python programming** for data analysis, machine learning, and workflow automation.
- Proficient in **C/C++** with experience in system-level programming and performance optimization.
- Experience with **TensorFlow and deep learning frameworks** for building, training, and deploying neural network models.
- Skilled in **end-to-end AI system development**, from data preprocessing to deployment and performance monitoring.
- Strong foundation in **data structures and algorithms**, enabling efficient and scalable solution design.
- Experience with **web data extraction and APIs**, supporting data-driven research and applications.
- Exposure to **distributed computing (Apache Spark)** and large-scale data processing.
- Applied AI expertise in **healthcare, image analysis, and intelligent systems**, with focus on real-world problem solving.

Academic Publications

Self-attention–guided EfficientAttentionNet-B0 for robust and automated tuberculosis detection from chest X-ray images. Array (Q1, 2026)
***Muhammad Zaman**, Zia ur Rehman, Saira Moin U Din, Adnan Akhunzada.*

Integrative Deep Diagnostic Architecture: A Convolutional Variational Attention Transformer for Personalized and Multi-Strategy Lung Nodule Detection. IEEE Access (Q1, 2026)
*Zia Ur Rehman, Juanjuan Zhao, **Muhammad Zaman** Corresponding Author, Qiang Yan, Aoyu Li, Adnan Akhunzada.*

Attention-Enhanced Deep Learning Framework for Multiclass Pulmonary Disease Diagnosis Using Chest X-Ray Imaging. (Q1, 2026)
*J Zhao, **Muhammad Zaman** (Corresponding Author), Q Yan, A Li, A Akhunzada.*

BroadAttNet: Attention-Driven Micro-Expression Recognition. CMC–Computers, Materials & Continua (Q1, 2026)
*Hafiz Khizer bin Talib, Yanlong Cao, **Muhammad Zaman** Corresponding Author, Sharifah Sakinah Syed Ahmad, Nikola Ivkovic, Mario Konecki, and Adnan Akhunzada.*

Cascaded Cross-Attention Vision Transformers with Wavelet-Based Encodings for DeepFake Detection. International Journal of Computer Vision (Q1, 2026)
*Polamarasetti, Anand, Shakeel Ahmad, **Muhammad Zaman** Corresponding Author, Rahiel Ahmad, Aftab Hussain, and Adnan Akhunzada.*

SepViT: A Dual-Path Transformer-Convolution Framework for Apex Frame-Based Microexpression Recognition. IEEE Transactions on Computational Social Systems (Q1, 2026)
*HKB Talib, Y Cao, **Muhammad Zaman** Corresponding Author, K Xu, A Akhunzada.*

Flight into the Future: A Holistic Review of AI Trends, Vision, and Challenges in Drone Technology.

Artificial Intelligence Review (Q1, 2025)
S Ahmad, R Ahmad, AS Al-Shamayleh, D Nimma, ***Muhammad Zaman*** *Corresponding Author, N Ivković.*

Parameter Estimation of Weibull Distribution Using Constrained Search Space: An Application to Elevator Maintenance. Machines (Q2, 2025)
K Ahmed, H Liu, L Ke, RT Mushtaq, ***Muhammad Zaman*** *Corresponding Author, A Akhunzada.*

Data-Efficient Wheat Disease Detection Using Shifted Window Transformer: Enhancing Accuracy, Sustainability, and Global Food Security. IEEE Transactions on Consumer Electronics (Q1, 2025)
M Khubaib, T Kehkashan, M Abdelhaq, MA Khan, ***Muhammad Zaman*** *Corresponding Author, I Ashraf.*

Error-Corrected Quantum CNN for Gender-Based Micro-Expression Analysis in Affective Computing. ICHORA (2025)
HKB Talib, K Xu, Y Cao, ***Muhammad Zaman*** *Corresponding Author, MH Toqeer, A Akhunzada.*

Early Detection of Diabetic Foot Ulcers through CoAtNet Model: A Scalable Solution for Healthcare Integration. ICHORA (2025)
MA Khan, ***Muhammad Zaman*** *Corresponding Author, T Kehkashan, N Ivković, V Uroš, A Akhunzada.*

A Novel Multi-Level Perceptron for Accurate Heart Stroke Diagnosis. BIT & CS Journal (2025)
Muhammad Zaman *Corresponding Author, M Khubaib, T Kehkashan, A Zahoor, N Shahbaz, S Shoukat.*

Micro-Expression Recognition Using Convolutional Variational Attention Transformer (ConVAT) with Multihead Attention Mechanism. IEEE Access (Q1, 2025)
HKB Talib, K Xu, Y Cao, YP Xu, Z Xu, ***Muhammad Zaman*** *Corresponding Author, A Akhunzada.*

Convolutional Attention-Based Mechanism for Facial Microexpression Recognition. IEEE Access (Q1, 2025)
HKB Talib, K Xu, Y Cao, YP Xu, Z Xu, ***Muhammad Zaman*** *Corresponding Author, A Akhunzada.*

Efficient Detection of Knee Osteoporosis Using Swin Transformer on X-Ray Images. ICCA (2024)
R Sultana, ***Muhammad Zaman*** *Corresponding Author, N Ivković, K Cengiz, I Rafaqat, A Akhunzada.*

Enhanced Pediatric Pneumonia Diagnosis with Chest X-Ray Using Deep Attention Mechanism. IEEE/ACM BDCAT (2024)
Muhammad Zaman *Corresponding Author, T Fatima, S Hameed, S Haider, A Akhunzada, M Azeem.*

Optimizing Diabetic Foot Ulcer Detection: Leveraging SSD and YOLO Architectures. IEEE Smart Communities (2024)
M Azeem, ***Muhammad Zaman*** *Corresponding Author, A Akhunzada, T Kehkashan, I Ashraf, A Rehman.*

Enhancing Micro-Expression Recognition with Broadbent Attention Mechanism: A High-Performance Approach for Emotion Detection. IEEE Smart Communities (2024)
I Ashraf, T Kehkashan, ***Muhammad Zaman*** *Corresponding Author, A Akhunzada, HK Talib, Y Cao.*

EQCNN: Enhanced Remote Sensing Imagery Classification with Circuit-Based Error-Corrected Quantum Convolutional Neural Networks. DICTA (2024)
Muhammad Zaman *Corresponding Author, T Kehkashan, A Akhunzada, H Alaidaros, M Uddin, M Azeem.*

Enhanced and Interpretable Brain Stroke Detection with High Accuracy Using ViT B-16 Model. ICIC (2024)
M Hamza, ***Muhammad Zaman*** *Corresponding Author, T Kehkashan, F Akbar, K Munir, A Shehzad.*

Enhanced Sign Language Detection with Deep CNN: Achieving Accuracy in Hand Gesture Recognition. ICIC (2024)
A Rehman, ***Muhammad Zaman*** *Corresponding Author, T Kehkashan, F Akbar, M Hamza, RA Riaz.*

A Hybrid Approach for Tuberculosis Detection Using Convolutional Attention Networks and XGBoost. ICIC (2024)
T Kehkashan, ***Muhammad Zaman*** *Corresponding Author, F Akbar, A Hussain, K Munir.*

Ensemble Learning for Breast Cancer Detection: A High-Accuracy Model Integrating Clinical Insights. ICIC (2024)
M Abdullah, T Kehkashan, ***Muhammad Zaman*** *Corresponding Author, F Akbar, RA Riaz, M Hamza.*

An Efficient Arthritis Diagnostic Model Using CoAtNet by Integrating Convolutional and Attention Mechanisms for Enhanced Precision. ICIC (2024)
A Shehzad, T Kehkashan, ***Muhammad Zaman*** *Corresponding Author, F Akbar, M Mohsin, AR Shoukat.*

Across the Spectrum: An In-Depth Review of AI-Based Models for Phishing Detection. IEEE Open Journal of the Communications Society (Q1, 2024)
S Ahmad, ***Muhammad Zaman*** *Corresponding Author, AS Al-Shamayleh, R Ahmad, SIM Abdulhamid.*

Citation Metrics: 164 citations (since 2021), h-index: 7, i10-index: 3

List of Graduates Supervised

I have supervised several **Master's by Research** students, contributing to their academic and research development across diverse domains in computer science and artificial intelligence.

- Iqra Rafaqat (Completed)
- Sana Hameed (Completed)
- Shakeel Ahmed (Completed)
- Muhammad Zubair (Completed)
- Muhammad Khubaib (Completed)
- Rahat Sultana (Completed)
- Tahseen Fatima (Completed)
- Hanzala Toqeer (Completed)
- Abuzar Ghafari (Completed)
- Nazish Huda (Completed)
- Aqsa Mahreen (Completed)
- Shahid Mahmood (Completed)
- Farhat Jabeen (Completed)
- Hina Ashraf (Completed)
- Ramsha Khaliq (Completed)

- Muhammad Jahanzaib **(In Progress)**
- Junaid Hamza **(In Progress)**
- Ali Zulfiqar **(In Progress)**
- Muhammad Arslaan **(In Progress)**
- Muhammad Bilal **(In Progress)**
- Hafiz Qamar Zaman **(In Progress)**
- Kaynat Rafaqat **(In Progress)**

Thesis Supervised

I have supervised several **Master's by Research** students. The supervised research primarily focuses on **Artificial Intelligence, Computer Vision, Explainable AI, and Medical Image Analysis**, contributing to both theoretical advancements and real-world applications.

1. A Generalized Explainable Deep Learning Framework for Interpretable and High-Fidelity Medical Image-Based Disease Diagnosis
2. A Transformer-Driven Hierarchical Architecture for Fine-Grained Micro-Expression Recognition in Affective Computing Systems
3. An Attention-Augmented Deep Neural Framework for Robust Multiclass Pulmonary Disease Detection Using Radiographic Imaging
4. Data-Efficient Deep Learning Paradigms for Scalable and Accurate Medical Image Classification under Limited Data Constraints
5. A Hybrid CNN-Transformer Architecture for High-Precision Brain Tumor Detection and Localization Using MRI Imaging Modalities
6. An Intelligent Deep Learning System for Automated Detection and Severity Assessment of Diabetic Foot Ulcers
7. A Vision Transformer-Based Framework for Accurate Osteoporosis Detection and Analysis from X-Ray Imaging Data
8. Explainable Artificial Intelligence Models for Transparent and Interpretable Clinical Decision Support Systems
9. A Deep Learning-Based End-to-End Framework for Automated Sign Language Recognition and Human-Computer Interaction
10. A Multi-Modal Deep Learning Architecture for Integrated Healthcare Data Fusion and Predictive Disease Modeling
11. Lightweight and Computationally Efficient Neural Network Architectures for Real-Time Medical Image Diagnostics
12. An AI-Driven Predictive Framework for Early Detection of Cardiovascular Diseases Using Multi-Source Clinical and Imaging Data

13. Quantum-Inspired Deep Learning Architectures for Enhanced Image Classification and Pattern Recognition
14. Deep Attention-Based Neural Architectures for Robust Emotion Recognition and Micro-Expression Analysis
15. A Scalable Artificial Intelligence Framework for Intelligent and Adaptive Healthcare Monitoring Systems
16. Explainable Neural Network Models for Reliable and Transparent Disease Classification in Clinical Applications
17. Transfer Learning-Driven Deep Models for Data-Constrained Medical Imaging Applications and Diagnostic Systems
18. An AI-Driven Diagnostic Framework for Automated Tuberculosis Detection Using Chest X-Ray Imaging
19. Transformer-Based Vision Models for Intelligent and Data-Efficient Crop Disease Detection in Smart Agriculture
20. Advanced Deep Learning-Based Image Segmentation Techniques for Precision Biomedical Image Analysis
21. Federated Learning-Based Architectures for Privacy-Preserving and Distributed Medical Artificial Intelligence Systems
22. Deep Learning-Based Anomaly Detection Frameworks for Early Identification of Irregularities in Medical Imaging Systems
23. Multi-Scale Feature Representation and Learning Framework for High-Accuracy and Robust Disease Diagnosis
24. An AI-Based Framework for Early and Accurate Stroke Detection Using Advanced Neuroimaging Data Analysis
25. Hybrid Deep Learning Architectures for High-Accuracy Breast Cancer Detection and Classification in Clinical Imaging

Various projects are on-going and will be updated upon successful completion.

Projects Supervised

I have supervised a diverse range of undergraduate Final Year Projects (FYPs) in the domains of **Artificial Intelligence, Computer Vision, Data Science, and Intelligent Software Systems**. These projects reflect strong integration of theoretical knowledge with practical implementation, focusing on real-world problem solving and innovation-driven development.

1. NeuroScan: Intelligent Brain Tumor Detection System
2. RetinaGuard: Diabetic Retinopathy Detection System
3. CancerVision: AI-Powered Cancer Detection System
4. VisionTrack: Real-Time Object Detection System
5. CrowdWatch: Intelligent Crowd Analysis System
6. SignAI: Sign Language Recognition System
7. FraudShield: Fraud Detection System
8. AgroAI: Smart Agriculture Monitoring System
9. MentorMeister: Online Learning Management System
10. ThoraxScan: Intelligent Chest Disease Detection System
11. Cogniosis: Smart Application for Brain Disease Detection
12. Rentify: Android Application for Property Rentals

Various additional projects are currently under supervision and will be incorporated upon completion.

Commercial Projects Running

- **Lead Gen AI Web Application**
  Role: Supervisor | Team Project (Completed)
  Led and supervised an **AI-powered lead generation system** that automates data scraping, lead extraction, and targeted email outreach, improving efficiency and conversion rates.
- **Auraq Publications Mobile App**
  Role: Supervisor | Team Project (Completed)
  Supervised an **AI-driven publishing platform** with Firebase backend, real-time data synchronization, workflow automation, and intelligent user interaction.

- **Blogging Website**
  Role: Supervisor | Team Project (Completed)
  Supervised a **scalable CMS platform** with secure access, modular design, and AI-driven content optimization.
- **Glamour Gems (E-commerce Platform)**
  Role: Supervisor | Team Project (Completed)
  Supervised development of a **full-stack e-commerce platform** using Node.js and MySQL with real-time inventory tracking, event-driven architecture, and optimized transaction management.
- **E-commerce Platform Website**
  Role: Supervisor | Team Project (Completed)
  Supervised development of a **secure, scalable e-commerce platform** with modular architecture, optimized backend services, and encrypted transaction processing.
- **Job Prep AI**
  Role: Supervisor | Team Project **(In Progress)**
  Supervised development of an **AI-driven career assistance platform** using NLP for resume analysis, adaptive interview simulation, and personalized skill recommendations.
- **Salon Management App**
  Role: Supervisor | Team Project **(In Progress)**
  Supervised development of a **multi-tenant salon management system** with real-time scheduling optimization, centralized architecture, and data-driven workflow automation.
- **Fantasy App (Handmade Marketplace)**
  Role: Supervisor | Team Project **(In Progress)**
  Led the development of a **Flutter-based cross-platform mobile marketplace** featuring a modular architecture, real-time transaction handling, and tailored recommendation capabilities.
- **Learn Quran Web App**
  Role: Supervisor | Team Project **(In Progress)**
  Supervised development of an **intelligent e-learning platform** with adaptive learning, progress tracking, and data-driven personalization.
- **Inventory Management & POS System**
  Role: Supervisor | Team Project **(In Progress)**
  Led design of a **MERN-based inventory and POS system** with real-time data synchronization, scalable architecture, and analytics for demand forecasting and decision support.
- **Digital Marketing Project**
  Role: Supervisor | Team Project **(In Progress)**
  Supervised development of a **data-driven digital marketing framework** with multi-channel integration, user behavior analytics, and predictive campaign optimization.
- **Inspectify (Home Maintenance Services App)**
  Role: Supervisor | Team Project **(In Progress)**
  Supervised design of a **service marketplace platform** with real-time booking, provider verification, and intelligent scheduling and routing optimization.
- **Construction Website Prototype**
  Role: Supervisor | Team Project **(In Progress)**
  Supervised design of a **web-based construction management system** with modular architecture, workflow optimization, and user-centric interface.
- **Hive: E-commerce Platform**
  Role: Supervisor | Team Project **(In Progress)**
  Supervised development of a **student-centric e-commerce platform** with verification, reputation systems, and scalable marketplace architecture.
- **Global Soft Management System**
  Role: Supervisor | Team Project **(In Progress)**
  Supervised design of an **integrated fuel management system** with real-time inventory synchronization, transaction processing, and analytics for sales forecasting and reporting.
- **RestoPOS – Restaurant Management System**
  Role: Supervisor | Team Project **(In Progress)**
  Supervised design of a **cloud-based restaurant management system** (Laravel, React) with real-time order processing, dynamic table management, and event-driven architecture.
- **Food Delivery Website**
  Role: Supervisor | Team Project **(In Progress)**
  Supervised development of a **food delivery platform** with real-time tracking, secure payments, and

intelligent routing for optimized delivery.

- **PulseFit (AI Fitness Platform)**
  Role: Supervisor | Team Project **(In Progress)**
  Supervised development of an **AI-driven fitness platform** with personalized training, recommendation systems, and real-time coaching via conversational AI.
- **Eduverse ERP Portal**
  Role: Supervisor | Team Project **(In Progress)**
  Supervised development of a **web-based ERP system** with AI-driven features including automated question generation and plagiarism detection for academic workflows.
- **School Management System**
  Role: Supervisor | Team Project (Planning)
  Supervised development of a **web-based school management system** with role-based access, workflow automation, and analytics for academic performance monitoring.
- **Tour & Travel Booking System**
  Role: Supervisor (Planning)
  Supervised development of a **travel booking platform** with real-time availability, unified service integration, and personalized recommendation features.
- **Shahar Bazar (Hyperlocal Marketplace)**
  Role: Supervisor | Startup Project (Planning)
  Supervised development of a **hyperlocal marketplace platform** with location-based matching, real-time transactions, and personalized recommendations.
- **POS Prime Tires**
  Role: Supervisor | Team Project (Planning)
  Supervised development of a **POS system for tire retail** with real-time inventory synchronization, transaction processing, and analytics for sales forecasting.
- **Professional Singing Events Booking Website**
  Role: Supervisor (Planning)
  Supervised development of a **performer booking platform** with intelligent matching, scheduling, and reputation-based ranking.
- **Custom Employee Management System**
  Role: Supervisor (Planning)
  Supervised design of an **enterprise HR management system** with RBAC security, audit trails, and automated workflow orchestration.

Areas of Expertise & Research Interests

- **Machine Learning:** Expertise in supervised, unsupervised, and semi-supervised learning paradigms, including model design, optimization, and performance evaluation. Focus on developing scalable and data-driven solutions for predictive analytics and intelligent decision-making systems.
- **Deep Learning:** Advanced knowledge of neural network architectures such as **Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), Transformers, and Attention Mechanisms**. Experience in designing high-performance models for complex tasks in vision, healthcare, and pattern recognition.
- **Computer Vision:** Specialized in visual data analysis, including object detection, image classification, segmentation, and video understanding. Application of deep learning techniques for real-world problems such as medical imaging, surveillance, and intelligent systems.
- **Image Processing:** Strong foundation in digital image processing techniques, including filtering, enhancement, feature extraction, and transformation. Integration of classical image processing methods with modern deep learning approaches for improved visual analysis.
- **Pattern Recognition:** Expertise in identifying and modeling patterns in complex datasets using statistical and machine learning techniques. Focus on feature representation, classification, clustering, and anomaly detection in high-dimensional data.
- **Reinforcement Learning:** Knowledge of decision-making frameworks based on reward-driven learning, including Markov Decision Processes (MDPs), policy optimization, and adaptive control systems for intelligent automation.
- **Quantum Machine Learning:** Emerging research interest in integrating quantum computing paradigms with machine learning models, including quantum-inspired algorithms and quantum neural networks for enhanced computational efficiency and problem-solving capabilities.

Professional Memberships and Research Affiliations

Institute of Electrical and Electronics Engineers (IEEE) — Member (Since 2023)
Member ID: 100129479 | Region 10 (Asia-Pacific) | Lahore Section | Faisalabad Subsection | Status: Active

- Active **IEEE member** (since 2023), engaged in global research networks, conferences, and scholarly activities in AI and computing.
- Member of **IEEE Young Professionals** and multiple societies (Computational Intelligence, Computer Society, Biometrics, Systems), contributing to interdisciplinary research in AI, systems, and intelligent technologies.
- Participate in **IEEE SIGHT and Consultants Network**, promoting AI-driven solutions for societal impact and industry-academia collaboration.
- Actively contribute to **research dissemination, academic networking, and international collaboration** through IEEE platforms.
- Engage in continuous professional development via **IEEE workshops, webinars, and training programs**.
- Integrate IEEE resources into **teaching, supervision, and research mentoring**, fostering a globally aligned academic environment.

Journal Referee Services & Editorial Contributions

I actively contribute to the global research community as a peer reviewer for several **high-impact, internationally recognized journals**, supporting the dissemination of high-quality scientific knowledge and ensuring adherence to rigorous scholarly standards. My reviewing activities focus on **Artificial Intelligence, Computer Vision, Machine Learning, Network Security, and Intelligent Systems**.

- **IEEE Transactions on Information Forensics and Security (TIFS)**
- **Computer Networks (COMNET), Elsevier**
- **Ad Hoc Networks (ADHOC), Elsevier**
- **IEEE Transactions on Vehicular Technology (TVT)**
- **Journal of Network and Computer Applications (JNCA), Elsevier**
- **IEEE Vehicular Technology Magazine**
- **IEEE Transactions on Intelligent Transportation Systems (T-TIS)**
- **IEEE Transactions on Intelligent Transportation Systems Magazine**
- **IEEE Transactions on Dependable and Secure Computing (TDSC)**
- **IEEE Transactions on Industrial Electronics (TIE)**
- **IEEE Communications Magazine**
- **IEEE Communications Letters**
- **Computer Communications, Elsevier**
- **Information Sciences (INS), Elsevier**
- **Information Fusion (INFFUS), Elsevier**
- **Telecommunication Systems (TELS), Springer**
- **Journal of Parallel and Distributed Computing, Elsevier**
- **IET Security**
- **Wireless Personal Communications (WIRE), Springer**
- **Journal of Network and System Management (JONS), Springer**
- **International Journal of Distributed Sensor Networks (IJDSN)**

*Through these roles, I contribute to maintaining **methodological rigor, research integrity, and scientific quality**, while engaging with cutting-edge advancements across interdisciplinary domains in computing and intelligent systems.*

Conferences and Seminars

- **Member — Technical Program Committee — 2026 International Conference on Applied Mathematics, Modeling in Engineering (AMME 2026)**
  December 28–29, 2026 • Changsha, China
- **Glob-Fist 2021 — Minhaj University Lahore (May 29–30, 2021)**
  Participated in an international academic conference focusing on emerging trends in **computer science, artificial intelligence, and interdisciplinary technologies**. Engaged with researchers and practitioners to explore advancements in intelligent systems, data-driven methodologies, and innovative computational frameworks, thereby enhancing research perspective and academic networking.
- **Preserving Our Planet: Understanding and Protecting the Environment — Superior University,**

Sargodha (November 2, 2024)
Attended a seminar addressing critical issues related to **environmental sustainability, climate resilience, and ecological conservation**. Explored the role of technology and policy in mitigating environmental challenges, with emphasis on integrating **data-driven and AI-based approaches** for sustainable development.

- **Effect of Climate Change on Human Health — Superior University, Sargodha (November 3, 2024)**
  Participated in an interdisciplinary seminar examining the **impact of climate change on public health**, including emerging health risks, epidemiological patterns, and preventive strategies. Gained insights into how computational models and data analytics can contribute to **healthcare forecasting and risk assessment**.
- **Crafting Your Masterpiece: Strategies for Writing a Compelling Thesis — Superior University, Sargodha (November 4, 2024)**
  Attended a professional development workshop focused on **advanced academic writing, research structuring, and scholarly communication**. Enhanced expertise in **thesis formulation, argument development, literature synthesis, and publication-quality writing**, supporting effective dissemination of research outcomes.
- **IEEE MINI POCO — University of Central Punjab, Lahore (November 23, 2024)**
  Participated in an IEEE-affiliated technical event emphasizing **innovation in computing, emerging technologies, and professional networking**. Engaged with the IEEE community to explore advancements in **AI, software systems, and engineering practices**, strengthening professional connections and exposure to industry-relevant developments.

Professional Training and Development

- **Social Entrepreneurial Program — Superior University Lahore (January 6–10, 2025)**
  Completed an intensive training program focused on **social entrepreneurship, innovation management, and impact-driven project development**. Acquired advanced understanding of **design thinking, problem identification, and solution-oriented strategies** aimed at addressing real-world societal challenges. Engaged in collaborative activities emphasizing **business model development, sustainability frameworks, and value creation**, fostering the ability to translate academic knowledge into socially impactful and scalable initiatives.
- **Social Entrepreneurial Program-II — Superior University Lahore (July 14–18, 2025)**
  Participated in an advanced-level continuation of the Social Entrepreneurial Program, with emphasis on **venture scaling, strategic innovation, and entrepreneurial leadership**. Developed competencies in **project incubation, resource optimization, and impact assessment**, while refining skills in pitching, stakeholder engagement, and interdisciplinary collaboration. Strengthened the capacity to integrate **technology-driven solutions, including AI-based approaches**, into sustainable entrepreneurial ventures aligned with global development goals.

Language Proficiency

- **Punjabi** — Native proficiency
  Demonstrates full professional and cultural fluency as a mother tongue, enabling effective communication, contextual understanding, and engagement in diverse social and academic environments.
- **Urdu** — C2 (Proficient User)
  Exhibits near-native command across all competencies, including **listening, reading, writing, spoken production, and spoken interaction**. Capable of engaging in complex academic discussions, formal communication, and professional discourse with high linguistic accuracy and clarity.
- **English** — C1 (Advanced Proficient User)
  Possesses advanced proficiency in **listening, reading, writing, and verbal communication**, enabling effective participation in academic writing, research dissemination, teaching, and international collaboration. Demonstrates strong capability in producing structured scholarly content and engaging in technical discussions within global academic contexts.

Proficiency Levels: A1–A2 (Basic User), B1–B2 (Independent User), C1–C2 (Proficient User)